Trigonometry & Polar Coordinates

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Convert the polar coordinates $\left(6, \frac{5\pi}{4}\right)$ into rectangular Cartesian coordinates (x, y).

Step 1: Apply coordinate conversion formulas: $x = r\cos\theta = 6\cos\left(\frac{5\pi}{4}\right) = 6\left(-\frac{\sqrt{2}}{2}\right) = -3\sqrt{2}$ Step 2: Compute y: $y = r\sin\theta = 6\sin\left(\frac{5\pi}{4}\right) = 6\left(-\frac{\sqrt{2}}{2}\right) = -3\sqrt{2}$ Thus the rectangular coordinates are $\boldsymbol{\left(-3\sqrt{2}, -3\sqrt{2}\right)}$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Convert the Cartesian equation $x^2 + (y-4)^2 = 16$ into a polar equation relating r and θ.

Step 1: Expand the binomial: $x^2 + y^2 - 8y + 16 = 16 \Rightarrow x^2 + y^2 - 8y = 0$. Step 2: Substitute polar identities $x^2 + y^2 = r^2$ and $y = r\sin\theta$: $r^2 - 8r\sin\theta = 0 \Rightarrow r(r - 8\sin\theta) = 0$. Step 3: Divide by r for $req0$ (the origin is included at $\theta = 0$): $\mathbf{r = 8\sin\theta}$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Evaluate the trigonometric limit: $\lim_{x\to 0} \frac{\sin(5x)}{\tan(2x)}$

Step 1: Express $\tan(2x)$ in terms of sine and cosine: $\frac{\sin(5x)}{\tan(2x)} = \frac{\sin(5x)}{\frac{\sin(2x)}{\cos(2x)}} = \frac{\sin(5x)}{\sin(2x)} \cdot \cos(2x)$ Step 2: Multiply and divide by appropriate factors of x: $\frac{\sin(5x)}{5x} \cdot 5x \cdot \frac{2x}{\sin(2x)} \cdot \frac{1}{2x} \cdot \cos(2x) = \frac{\sin(5x)}{5x} \cdot \frac{1}{\frac{\sin(2x)}{2x}} \cdot \frac{5}{2} \cdot \cos(2x)$ Step 3: Evaluate limits using $\lim_{u\to 0} \frac{\sin u}{u} = 1$ and $\cos(0) = 1$.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the trigonometric equation on the interval $[0, 2\pi)$: $2\cos^2 x - 3\cos x + 1 = 0$

Step 1: Factor as a quadratic in $\cos x$: $(2\cos x - 1)(\cos x - 1) = 0$ Step 2: Solve each factor: 1. $2\cos x - 1 = 0 \Rightarrow \cos x = \frac{1}{2} \Rightarrow \mathbf{x = \frac{\pi}{3}, \quad \frac{5\pi}{3}}$. 2. $\cos x - 1 = 0 \Rightarrow \cos x = 1 \Rightarrow \mathbf{x = 0}$. All solutions in $[0, 2\pi)$: $\mathbf{0, \quad \frac{\pi}{3}, \quad \frac{5\pi}{3}}$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Evaluate the limit: $\lim_{x\to 0} \frac{\sqrt{1+x} - 1}{x}$

Step 1: Rationalize numerator using the conjugate $\sqrt{1+x} + 1$: $\lim_{x\to 0} \frac{(\sqrt{1+x} - 1)(\sqrt{1+x} + 1)}{x(\sqrt{1+x} + 1)} = \lim_{x\to 0} \frac{(1+x) - 1}{x(\sqrt{1+x} + 1)}$ Step 2: Cancel x and evaluate: $\lim_{x\to 0} \frac{x}{x(\sqrt{1+x} + 1)} = \lim_{x\to 0} \frac{1}{\sqrt{1+x} + 1} = \frac{1}{\sqrt{1+0} + 1} = \mathbf{\frac{1}{2}}$.

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6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the scalar projection of $\mathbf{u} = \langle 4, 3 \rangle$ onto $\mathbf{v} = \langle 12, 5 \rangle$.

Formula: $\text{comp}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|}$. Step 1: Compute $\mathbf{u} \cdot \mathbf{v}$: $\mathbf{u} \cdot \mathbf{v} = (4)(12) + (3)(5) = 48 + 15 = 63$. Step 2: Compute $|\mathbf{v}|$: $|\mathbf{v}| = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$. Step 3: Calculate scalar projection: $\text{comp}_{\mathbf{v}}\mathbf{u} = \mathbf{\frac{63}{13}}$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Evaluate the power of the imaginary unit: $i^{2026} + i^{2027} + i^{2028} + i^{2029}$

Step 1: Recall the periodicity of powers of i: Powers of i repeat cyclically with period 4: $i^1 = i$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$. The sum of any 4 consecutive integer powers of i is identically zero: $i^1 + i^2 + i^3 + i^4 = i - 1 - i + 1 = 0$. Step 2: Apply to the consecutive sequence: $i^{2026} + i^{2027} + i^{2028} + i^{2029} = (-1) + (-i) + (1) + (i) = \mathbf{0}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A constant force vector $\mathbf{F} = \langle 8, -3 \rangle$ (in newtons) moves an object along the displacement vector $\mathbf{d} = \langle 5, 2 \rangle$ (in meters). Calculate the work done by the force.

Definition of Work: $W = \mathbf{F} \cdot \mathbf{d}$. Step 1: Evaluate the dot product: $W = (8)(5) + (-3)(2) = 40 - 6 = \mathbf{34}$ J. Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Given that $\sin\theta = \frac{3}{5}$ with θ in Quadrant II, find the exact value of $\sin(2\theta)$.

Step 1: Find $\cos\theta$: In Quadrant II, cosine is negative: $\cos\theta = -\sqrt{1 - \sin^2\theta} = -\sqrt{1 - \left(\frac{3}{5}\right)^2} = -\sqrt{\frac{16}{25}} = -\frac{4}{5}$. Step 2: Apply double-angle formula: $\sin(2\theta) = 2\sin\theta\cos\theta = 2\left(\frac{3}{5}\right)\left(-\frac{4}{5}\right) = \mathbf{-\frac{24}{25}}$.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Evaluate the indeterminate limit: $\lim_{x \to 5} \frac{x^2 - 25}{x^2 - 8x + 15}$

Step 1: Direct substitution check: $\frac{5^2 - 25}{5^2 - 40 + 15} = \frac{0}{0}$ (indeterminate). Step 2: Factor numerator and denominator: $\frac{(x-5)(x+5)}{(x-5)(x-3)}$ Step 3: Cancel common factor and evaluate: $\lim_{x \to 5} \frac{x+5}{x-3} = \frac{5+5}{5-3} = \frac{10}{2} = \mathbf{5}$.

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In triangle ABC, $A = 45^\circ$, $B = 60^\circ$, and $a = 8$. Find the exact length of side b.

Step 1: Apply Law of Sines: $\frac{b}{\sin B} = \frac{a}{\sin A} \Rightarrow b = \frac{a\sin B}{\sin A}$ Step 2: Substitute exact values: $b = \frac{8\sin 60^\circ}{\sin 45^\circ} = \frac{8\left(\frac{\sqrt{3}}{2}\right)}{\frac{\sqrt{2}}{2}} = \frac{4\sqrt{3}}{\frac{\sqrt{2}}{2}} = \frac{8\sqrt{3}}{\sqrt{2}} = \mathbf{4\sqrt{6}}$.

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12 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Vectors $\mathbf{u} = \langle 3, -2, 5 \rangle$ and $\mathbf{v} = \langle 4, k, -2 \rangle$ are orthogonal in $\mathbb{R}^3$. Find the value of k.

Step 1: Orthogonality criterion: Two nonzero vectors are orthogonal if and only if their dot product equals 0: $\mathbf{u} \cdot \mathbf{v} = 0$ Step 2: Compute dot product: $(3)(4) + (-2)(k) + (5)(-2) = 0$ $12 - 2k - 10 = 0 \Rightarrow 2 - 2k = 0 \Rightarrow 2k = 2 \Rightarrow \mathbf{k = 1}$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Evaluate the limit at infinity: $\lim_{x \to \infty} \frac{5x^3 - 2x + 7}{2x^3 + 4x^2 - 1}$

Step 1: Compare degrees of numerator and denominator: Both numerator and denominator have degree 3. Step 2: Apply Theorem 6.1 (Ratio of leading coefficients): $\lim_{x \to \infty} \frac{5x^3 - 2x + 7}{2x^3 + 4x^2 - 1} = \frac{5}{2} = \mathbf{\frac{5}{2}}$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Given $A = (2, -1; 3, 4)$ and $B = (0, 5; -2, 1)$, compute $3A - 2B$.

Step 1: Compute scalar multiples: $3A = (6, -3; 9, 12)$, $2B = (0, 10; -4, 2)$ Step 2: Subtract entrywise: $3A - 2B = (6 - 0, -3 - 10; 9 - (-4), 12 - 2) = (\mathbf{6, -13; 13, 10})$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Advanced Functions & Equations

Use the Remainder Theorem to find the remainder when $P(x) = 3x^4 - 2x^3 + 5x - 7$ is divided by $x + 2$.

Remainder Theorem: The remainder upon division by $x - c$ is $P(c)$. Here $c = -2$: $P(-2) = 3(-2)^4 - 2(-2)^3 + 5(-2) - 7$ $= 3(16) - 2(-8) - 10 - 7 = 48 + 16 - 10 - 7 = 64 - 17 = \mathbf{47}$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 Limits

In triangle ABC, $a = 7$, $b = 5$, and $\cos C = \frac{1}{5}$. (a) Find the exact length of side c. (b) Calculate the exact value of $\sin^2 C$ and the area of triangle ABC.

Part (a): Find side c using Law of Cosines: $c^2 = a^2 + b^2 - 2ab\cos C$ $c^2 = 7^2 + 5^2 - 2(7)(5)\left(\frac{1}{5}\right) = 49 + 25 - 14 = 60 \Rightarrow \mathbf{c = \sqrt{60} = 2\sqrt{15}}$. Part (b): Find $\sin^2 C$ and Area: 1. Pythagorean identity: $\sin^2 C = 1 - \cos^2 C = 1 - \left(\frac{1}{5}\right)^2 = 1 - \frac{1}{25} = \mathbf{\frac{24}{25}}$. 2. Since C is an angle of a triangle, $\sin C > 0 \Rightarrow \sin C = \frac{\sqrt{24}}{5} = \frac{2\sqrt{6}}{5}$. 3. Area formula.

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17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Complex Numbers

Convert the polar equation $r = 6\cos\theta$ into Cartesian coordinates, identify the conic section, and state its center and radius.

Step 1: Multiply both sides by r: $r^2 = 6r\cos\theta$ Step 2: Substitute Cartesian conversion formulas $r^2 = x^2 + y^2$ and $x = r\cos\theta$: $x^2 + y^2 = 6x$ Step 3: Move all terms to one side and complete the square in x: $(x^2 - 6x) + y^2 = 0$ $(x-3)^2 - 9 + y^2 = 0 \Rightarrow \mathbf{(x-3)^2 + y^2 = 9}$. Conclusion: The equation represents a circle with center $\mathbf{(3, 0)}$ and radius $\mathbf{r = 3}$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Matrices & Vectors

A student spends 90 seconds on each of the first 20 questions of the 50-question AIST Math 2 exam. How much average time (in seconds) remains per question for the remaining 30 questions?

Total exam duration is $60 \text{ minutes} = 3600 \text{ seconds}$. The first 20 questions take $20 \times 90 = 1800 \text{ seconds}$, leaving $3600 - 1800 = 1800 \text{ seconds}$ for the remaining 30 questions. The remaining time per question is $\frac{1800}{30} = 60 \text{ seconds}$ (1 minute per question). Pacing discipline early on prevents a frantic finish.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Sequences & Series

Compute $(\sqrt{3} - i)^8$ using De Moivre's Theorem, expressing the result in standard rectangular form $a + bi$.

Step 1: Polar form of $z = \sqrt{3} - i$: Modulus: $r = \sqrt{(\sqrt{3})^2 + (-1)^2} = \sqrt{3+1} = 2$. Argument: In Quadrant IV with $\tan\theta = -\frac{1}{\sqrt{3}} \Rightarrow \theta = -\frac{\pi}{6}$. $z = 2\left(\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right)$. Step 2: Apply De Moivre's Theorem.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Advanced Functions & Equations

Evaluate the limit at infinity involving a square root in the numerator: $\lim_{x\to\infty} \frac{\sqrt{9x^2 + 4x}}{2x - 1}$

Step 1: Divide numerator and denominator by x (for $x > 0$, $x = \sqrt{x^2}$): $\frac{\sqrt{9x^2+4x}}{2x-1} = \frac{\frac{\sqrt{9x^2+4x}}{\sqrt{x^2}}}{\frac{2x-1}{x}} = \frac{\sqrt{\frac{9x^2}{x^2} + \frac{4x}{x^2}}}{2 - \frac{1}{x}} = \frac{\sqrt{9 + \frac{4}{x}}}{2 - \frac{1}{x}}$. Step 2: Take the limit as $x \to \infty$: Since $\frac{4}{x} \to 0$ and $\frac{1}{x} \to 0$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Express $2\sin(5x)\cos(3x)$ as a sum of trigonometric functions.

a. $\sin(8x) + \sin(2x)$ ✓

b. $\sin(8x) - \sin(2x)$

c. $\cos(8x) + \cos(2x)$

d. $\cos(8x) - \cos(2x)$

Using the product-to-sum identity $2\sin A\cos B = \sin(A + B) + \sin(A - B)$ gives $\sin(8x) + \sin(2x)$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Simplify the expression $\frac{\sin(3x) + \sin(x)}{\cos(3x) + \cos(x)}$.

a. $\tan(2x)$ ✓

b. $\tan(x)$

c. $\cot(2x)$

d. $\tan(4x)$

Using sum-to-product: numerator $= 2\sin(2x)\cos(x)$ and denominator $= 2\cos(2x)\cos(x)$. Dividing gives $\frac{\sin(2x)}{\cos(2x)} = \tan(2x)$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the exact value of $\cos(105^\circ) - \cos(15^\circ)$?

a. $-\frac{\sqrt{2}}{2}$ ✓

b. $\frac{\sqrt{2}}{2}$

c. $-\frac{\sqrt{6}}{2}$

d. $\frac{\sqrt{6}}{2}$

Sum-to-product: $-2\sin(60^\circ)\sin(45^\circ) = -2(\frac{\sqrt{3}}{2})(\frac{\sqrt{2}}{2}) = -\frac{\sqrt{6}}{2}$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What geometric curve is represented by the polar equation $r = 2 + 2\cos\theta$?

a. **Cardioid** ✓

b. Limaçon with inner loop

c. Three-petal rose

d. Circle

An equation of the form $r = a + a\cos\theta$ has equal coefficients ($a = b$), which represents a cardioid (heart-shaped curve).

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How many petals does the rose curve $r = 4\cos(3\theta)$ have?

a. 3 ✓

b. 6

c. 4

d. 8

For a rose curve $r = a\cos(n\theta)$, if n is odd, the curve has n petals. Here $n = 3$, so it has 3 petals.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert the polar point $(4, \frac{5\pi}{6})$ to rectangular coordinates (x, y).

a. $(-2\sqrt{3}, 2)$ ✓

b. $(2\sqrt{3}, -2)$

c. $(-2, 2\sqrt{3})$

d. $(2, -2\sqrt{3})$

$x = r\cos\theta = 4\cos(\frac{5\pi}{6}) = 4(-\frac{\sqrt{3}}{2}) = -2\sqrt{3}$. $y = r\sin\theta = 4\sin(\frac{5\pi}{6}) = 4(\frac{1}{2}) = 2$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert the rectangular point $(-3, -3)$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

a. $(3\sqrt{2}, \frac{5\pi}{4})$ ✓

b. $(3\sqrt{2}, \frac{3\pi}{4})$

c. $(6, \frac{5\pi}{4})$

d. $(3\sqrt{2}, \frac{7\pi}{4})$

$r = \sqrt{(-3)^2 + (-3)^2} = \sqrt{18} = 3\sqrt{2}$. Point is in Quadrant III: $\theta = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the Cartesian equation for the polar line $\theta = \frac{\pi}{3}$?

a. $y = \sqrt{3}x$ ✓

b. $y = \frac{\sqrt{3}}{3}x$

c. $y = x + \frac{\pi}{3}$

d. $x = \sqrt{3}y$

Take tangent of both sides: $\tan\theta = \tan(\frac{\pi}{3}) = \sqrt{3}$. Since $\tan\theta = \frac{y}{x}$, we have $\frac{y}{x} = \sqrt{3} \Rightarrow y = \sqrt{3}x$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find all solutions to $\sin(2x) = \cos(x)$ in the interval $[0, 2\pi)$.

a. $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$ ✓

b. $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$

c. $\frac{\pi}{3}, \frac{2\pi}{3}$

d. $\frac{\pi}{6}, \frac{5\pi}{6}$

$2\sin x\cos x - \cos x = 0 \Rightarrow \cos x(2\sin x - 1) = 0$. $\cos x = 0 \Rightarrow x = \frac{\pi}{2}, \frac{3\pi}{2}$. $\sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the range of $f(x) = 2 - 3\sin(4x)$?

a. $[-1, 5]$ ✓

b. $[-3, 3]$

c. $[-5, 1]$

d. $[2, 5]$

Since $-1 \le \sin(4x) \le 1$, we have $-3 \le -3\sin(4x) \le 3$. Adding 2 gives $-1 \le f(x) \le 5$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the standard (x, y)-coordinate plane, what are the rectangular coordinates of the point with polar coordinates $(r, \theta) = \left(6, \frac{5\pi}{6}\right)$?

a. $(-3, 3\sqrt{3})$

b. $(-3\sqrt{3}, 3)$ ✓

c. $(3\sqrt{3}, -3)$

d. $(-3\sqrt{3}, -3)$

To convert from polar coordinates (r, θ) to rectangular coordinates (x, y), use the conversion formulas: $x = r\cos\theta$ and $y = r\sin\theta$ Given $r = 6$ and $\theta = \frac{5\pi}{6}$: $x = 6\cos\left(\frac{5\pi}{6}\right) = 6\left(-\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}$ $y = 6\sin\left(\frac{5\pi}{6}\right) = 6\left(\frac{1}{2}\right) = 3$ Thus, the rectangular coordinates are $(-3\sqrt{3}, 3)$. - Choice A $(-3, 3\sqrt{3})$ results from inadvertently reversing the sine and c.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A point in the polar coordinate plane has coordinates $(r, \theta) = \left(4, \frac{5\pi}{6}\right)$. Which of the following gives the rectangular coordinates (x, y) of this point?

a. $(-2\sqrt{3}, 2)$ ✓

b. $(-2, 2\sqrt{3})$

c. $(2, -2\sqrt{3})$

d. $(2\sqrt{3}, -2)$

To convert polar coordinates (r, θ) to rectangular coordinates (x, y), use the conversion formulas: $x = r\cos\theta$ and $y = r\sin\theta$ Given $r = 4$ and $\theta = \frac{5\pi}{6}$: $x = 4\cos\left(\frac{5\pi}{6}\right) = 4\left(-\frac{\sqrt{3}}{2}\right) = -2\sqrt{3}$ $y = 4\sin\left(\frac{5\pi}{6}\right) = 4\left(\frac{1}{2}\right) = 2$ Thus, the rectangular coordinates are $(-2\sqrt{3}, 2)$. - Choice B is incorrect because it mistakenly swaps the values of.

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Convert the rectangular point $(-3.464, -2)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(4, 210^\circ)$ ✓

b. $(4, 30^\circ)$

c. $(4, 150^\circ)$

d. $(8, 210^\circ)$

$r = \sqrt{x^2 + y^2} \approx 4$; the reference angle and the quadrant of $(-3.464, -2)$ give $\theta = 210^\circ$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC, $A = 54^\circ$, $a = 10$, $B = 92^\circ$. Find b to the nearest tenth.

a. **12.4** ✓

b. 13.4

c. 11.9

d. 10

$\frac{10}{\sin 54^\circ} = \frac{b}{\sin 92^\circ} \Rightarrow b = \frac{10\sin 92^\circ}{\sin 54^\circ} \approx 12.4.$

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In the polar coordinate plane, a curve is defined by the equation $r = 6\cos\theta + 8\sin\theta$. Two rays emanating from the pole at angles $\theta = \alpha$ and $\theta = \alpha + \frac{\pi}{2}$ intersect the curve at points $PeqO$ and $QeqO$, respectively, where O denotes the pole. Assuming α is chosen such that both P and Q have positive radial coordinates, what is the maximum possible area of ΔOPQ?

a. **25** ✓

b. 24

c. 48

d. 50

We can solve this problem using either trigonometric synthesis or coordinate geometry. **Method 1: Trigonometric Synthesis** The radial coordinates of P and Q along their respective rays are: $r_1 = 6\cos\alpha + 8\sin\alpha$ $r_2 = 6\cos(\alpha + \frac{\pi}{2}) + 8\sin(\alpha + \frac{\pi}{2}) = -6\sin\alpha + 8\cos\alpha$ Since the angle between the two rays is $(\alpha + \frac{\pi}{2}) - \alpha = \frac{\pi}{2}$, ΔOPQ is a right-angled triangle at the pole O. Its are.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In the polar coordinate plane, points A and B lie on the curve $r = 4\sin(2\theta)$ such that their polar angles satisfy $0 < \theta_A < \theta_B < \frac{\pi}{2}$ and $\theta_B - \theta_A = \frac{\pi}{3}$. What is the maximum possible area of ΔOAB, where O is the origin?

a. $\sqrt{3}$ ✓

b. 2

c. $2\sqrt{3}$

d. $4\sqrt{3}$

In polar coordinates, the area of a triangle formed by the pole O and two points $A(r_A, \theta_A)$ and $B(r_B, \theta_B)$ is given by: $\text{Area} = \frac{1}{2} r_A r_B \sin(\theta_B - \theta_A)$ Given $\theta_B - \theta_A = \frac{\pi}{3}$, we have: $\text{Area} = \frac{1}{2} r_A r_B \sin(\frac{\pi}{3}) = \frac{\sqrt{3}}{4} r_A r_B$ For points on the curve $r = 4\sin(2\theta)$, since 00 and $r_B > 0$. Let $\theta_A = \alpha$; then $\theta_B = \alpha + \frac{\pi}{3}$, where.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the polar coordinate system, point A has polar coordinates (r_A, θ_A) lying on the curve $r = 6\sin\theta$, and point B has polar coordinates (r_B, θ_B) lying on the curve $r = 8\cos\theta$, with $r_A > 0$ and $r_B > 0$. If $\theta_A - \theta_B = \frac{\pi}{6}$, what is the maximum possible area of ΔOAB, where O is the pole (origin)?

a. 6

b. $6\sqrt{3}$

c. **9** ✓

d. 12

In polar coordinates, the area of a triangle with one vertex at the pole $O(0, 0)$ and the other two vertices at $A(r_A, \theta_A)$ and $B(r_B, \theta_B)$ is given by: $\text{Area}(\Delta OAB) = \frac{1}{2} r_A r_B \sin|\theta_A - \theta_B|$ We are given $\theta_A - \theta_B = \frac{\pi}{6}$, so $\sin|\theta_A - \theta_B| = \sin(\frac{\pi}{6}) = \frac{1}{2}$. Let $\theta_B = \theta$. Then $\theta_A = \theta + \frac{\pi}{6}$. Since point A lies on $r = 6\sin\theta$ and point B lies on.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 2.01 Matrices & Vectors**

A point has polar coordinates $(r, \theta) = \left(6, \frac{2\pi}{3}\right)$. What are the Cartesian coordinates (x, y) of this point?

a. $(-3, 3\sqrt{3})$ ✓

b. $(-3\sqrt{3}, 3)$

c. $(3, -3\sqrt{3})$

d. $(3\sqrt{3}, -3)$

To convert polar coordinates (r, θ) to Cartesian coordinates (x, y), use the conversion formulas: $x = r\cos\theta$ and $y = r\sin\theta$ Given $r = 6$ and $\theta = \frac{2\pi}{3}$: $x = 6\cos\left(\frac{2\pi}{3}\right) = 6\left(-\frac{1}{2}\right) = -3$ $y = 6\sin\left(\frac{2\pi}{3}\right) = 6\left(\frac{\sqrt{3}}{2}\right) = 3\sqrt{3}$ Thus, the Cartesian coordinates are $(-3, 3\sqrt{3})$. Common errors: - Interchanging x and y (using $x = r\sin\theta$ and $y = r\cos\theta$.

39 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 3.01 Sequences & Series**

A point in the polar coordinate system has coordinates $(r, \theta) = \left(6, \frac{5\pi}{6}\right)$. Which of the following represents the Cartesian coordinates (x, y) of this point?

a. $(-3\sqrt{3}, 3)$ ✓

b. $(-3, 3\sqrt{3})$

c. $(3\sqrt{3}, -3)$

d. $(-3\sqrt{3}, -3)$

To convert polar coordinates (r, θ) to Cartesian coordinates (x, y), use the conversion formulas: $x = r\cos\theta$ and $y = r\sin\theta$ Given $r = 6$ and $\theta = \frac{5\pi}{6}$: $x = 6\cos\left(\frac{5\pi}{6}\right) = 6\left(-\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}$ $y = 6\sin\left(\frac{5\pi}{6}\right) = 6\left(\frac{1}{2}\right) = 3$ Thus, the Cartesian coordinates of the point are $(-3\sqrt{3}, 3)$. Common errors include swapping the cosine and sine components (resulting in.

40 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 4.01 Advanced Functions & Equations**

If $\sin x = \frac{7}{25}$ and $\cos x = \frac{24}{25}$ (Quadrant I), find $\sin(2x)$.

a. $\frac{336}{625}$ ✓

b. $\frac{527}{625}$

c. $\frac{7}{25}$

d. $\frac{24}{25}$

$\sin 2x = 2\sin x\cos x = 2\left(\frac{7}{25}\right)\left(\frac{24}{25}\right) = \frac{336}{625}$.

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Limits

In which quadrant does the terminal side of 65° lie?

a. **Quadrant 1** ✓

b. Quadrant 2

c. Quadrant 3

d. Quadrant 4

65° falls between the boundary angles of Quadrant 1 (0°–90°).

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Complex Numbers

Convert $(r, \theta) = (2, 30^\circ)$ to rectangular coordinates.

a. $(3,\ 1)$ ✓

b. $(1,\ 3)$

c. $(-3,\ 1)$

d. $(3,\ -1)$

$x = 2\cos 30^\circ$, $y = 2\sin 30^\circ$, giving (\sqrt3,\ 1).

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Matrices & Vectors

In the xy-plane, let $P(t, t^2)$ with $t > 0$ be a point on the parabola $y = x^2$. The normal line to the parabola at P intersects the parabola again at a second point Q. Let T be the intersection point of the tangent lines to the parabola at P and Q. As $t \to 0^+$, what is the limiting value of the y-coordinate of T?

a. $-\frac{1}{2}$ ✓

b. $-\frac{1}{4}$

c. -1

d. 0

We synthesize differential calculus, coordinate geometry, and limits to determine the behavior of T as $t \to 0^+$. 1. **Equation of the Normal Line at P:** Since $y = x^2$, the derivative is $y' = 2x$. At point $P(t, t^2)$, the slope of the tangent line is $2t$. Therefore, the slope of the normal line is the negative reciprocal: $m_{\text{normal}} = -\frac{1}{2t}$ The equation of the normal line at P is: $y - t^2 = -\frac{1}{2t}(x - t)$ 2. **Finding the Second Intersection Point Q:** To find wher.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Sequences & Series

For real constants a and b, suppose that $\lim_{x\to 0} \frac{\sqrt{ax+b}-3}{x} = 2$. What is the value of $a+b$?

a. 12

b. 15

c. **21** ✓

d. 27

Step 1: Determine the value of b. As $x \to 0$, the denominator approaches 0. For the limit to exist and equal the finite real number 2, the numerator must also approach 0 to create an indeterminate form of $\frac{0}{0}$: $\lim_{x\to 0}\left(\sqrt{ax+b}-3\right) = \sqrt{b}-3 = 0 \Rightarrow \sqrt{b} = 3 \Rightarrow b = 9$. Step 2: Evaluate the limit in terms of a. Substitute $b = 9$ and rationalize the numerator by multiplying by the conjugate.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Advanced Functions & Equations

For each real number $t \in (-2, 2)$, the cubic equation $x^3 - 3x + t = 0$ has three distinct real roots, denoted in increasing order by $x_1(t) < x_2(t) < x_3(t)$. What is the value of the limit $\lim_{t\to 2^-} \frac{x_3(t)-x_2(t)}{\sqrt{2-t}}$?

a. $\frac{2\sqrt{3}}{3}$ ✓

b. $\sqrt{3}$

c. $\frac{\sqrt{3}}{2}$

d. $\frac{4}{3}$

To evaluate the limit, we analyze the behavior of the roots as $t \to 2^-$ using polynomial factorization and Vieta's formulas. **Step 1: Determine the limiting values of the roots.** When $t = 2$, the equation becomes: $x^3 - 3x + 2 = (x-1)^2(x+2) = 0$ The roots are -2 (simple root) and 1 (double root). Since $x_1(t)x_2(t)$, the difference $x_3(t) - x_2(t)$ is strictly positive. Taking the square root gives.

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